CSE-214
Assignment on Load Testing with JMeter

Submission Deadline: 8 April, 2026

Apache JMeter is a versatile, open-source tool used to simulate high user loads and measure application responsiveness. In this assignment, you will evaluate the performance of a public API and a locally hosted Node.js server. You will analyze key metrics such as response times, throughput, and error rates under different load conditions.

Test Environment & Target

You can choose any two websites as you like to complete the assignment. Just mention the web addresses you used in your assignment in the word file.

Test Requirements

- **Load Profiles:** Run tests for **50 samples** and **100 samples**.
- **Ramp-up Time:** Set to **100 seconds** for all tests.
- **Logic Controllers:** Use **Recording Controllers** for each application to organize your samples.
- **Test Scripting:** Add an **HTTP(S) Test Script Recorder** to your test plan to capture the requests for the public API.
- **Assertions:** Configure a **Duration Assertion**.
    - *Constraint:* Set the threshold so that you observe **both successful and failed events** (e.g., if average response time is 200ms, set the assertion to 150ms to trigger some "failures").

Make necessary assumptions, perform necessary setups and set necessary listeners, assertions, logic controller and other tools for successful execution of testing.

Deliverables

1. **yourID_Assessment.docx**: A report containing:
    - **(a)** Comparison tables of **Shortest, Longest, and Average** execution times for 50 vs. 100 samples across both targets.
    - **(b)** Comparison of **Average Throughput**, **Data Reception (KB/sec)**, and **Data Transmission (KB/sec)**.
2. **yourID_testPlan.jmx**: Your exported JMeter Test Plan.
3. **yourID_result.csv**: The raw aggregate results from your test runs.
4. **yourID_html.zip**: The generated JMeter HTML Dashboard Report.